

The dirt on biochar

Melbourne Polytechnic turf management teacher and former AGCSATech agronomist Steve Tuckett looks at the use of biochar as a sports turf profile amendment.

Public awareness has increased regarding environmental issues which has led to more scrutiny by regulatory authorities. These issues include pesticide use, leaching of fertilisers and pesticides into water bodies and ground water and the factors that contribute to global warming.

In response to this, sports turf managers are always looking at ways to reduce inputs, including pesticide and fertiliser use, on their surfaces. Improving soil health that leads to improved plant health is one way to do that as a healthy plant requires less inputs of fertiliser and pesticides to produce a quality turf surface.

There are many success stories from using organic and high-carbon products to reduce reliance on chemicals. Recently, biochar has started to become popular and a lot has been made of the benefits of it by those companies selling the product, but little of the potential issues of using it as an amendment in sports turf rootzones.

Biochar does have benefits for the soil's food web, such as increasing the soils cation exchange capacity (CEC) and therefore increasing nutrient retention (Glaser et al., 2002; Lehmann et al., 2003). According to various researchers (Karhu et al., 2011; Peng et al. 2011; Ding et al., 2010; and Basso et al. 2013), biochar can increase moisture retention (although the results vary widely), so it could be an effective amendment to be used in rootzones with high infiltration rate. Biochar is similar to zeolite as it has a honeycomb structure giving it a very high surface area.

A common theme for a sales pitch for biochar is… "Biochar is a stable form of soil carbon with a naturally porous structure which improves aeration, water-holding capacity and nutrient retention of soils and acts as a refuge for beneficial soil microbiology."

To improve the nutrient and water retention in sandy rootzones, peat moss is commonly incorporated (Waddington, 1992). However, over time the decomposition of peat moss adversely affects its positive impacts on the rootzone as it breaks down into fine particles and reduces water infiltration rates and thus losing some of its nutrient retention capacity in a sand-based system. (Bigelow et al., 2004). According to Brockhoff (2010), biochar would seem to be a viable alternative as it may have similar benefits to peat moss while being less prone to decomposition.

That all sounds positive, but the question is if we amend sports turf profiles with biochar where drainage is key, what will be the long-term effects of this practice? Biochar is not biodegradable so once it is in the rootzone



Biochar is a product produced by heating organic materials, such as wood, crop waste and manures, in a low oxygen environment. The result is a porous, charcoal-like biochar rich in carbon content that can effectively capture carbon and lock the carbon into the soil

PHOTO: SANGSIRIPECH/STOCK.ADOBE.COM

it is there permanently and any reduction in drainage rate will need a lot of remedial work to rectify. As well, there are many different types of biochar, so you need to do your research to determine which one is the most suitable for your situation.

WHAT IS BIOCHAR?

Biochar is a product produced by heating organic materials, such as wood, crop waste and manures, in a low oxygen environment. The heating process is called 'pyrolysis', where the organic materials are placed into a special oven where heating occurs in the presence of little or no oxygen. The result is a porous, charcoal-like biochar rich in carbon content that can effectively capture carbon and lock the carbon into the soil.

Many turf managers would be familiar with activated carbon which has been used for remediation of areas where chemical spills have occurred. Activated carbon and charcoal are also produced through the heating of organic materials via pyrolysis, so what is the difference, if any? Put simply, it's about the end use as biochar is used as a soil amendment, charcoal is used as a fuel and activated carbon for filtration, remediation and purification.

The temperature and duration of the pyrolysis process and the original ingredients will ultimately determine the biochar's properties. Biochar materials possess different particle size characteristics depending on the temperature and burn times of the pyrolysis process. Dall'Ora et al. (2008) note that fast pyrolysis occurs at high temperatures with a short burn period and produces a fine-textured biochar (< .002mm diameter particle size). Conversely, slow pyrolysis produces a coarse-textured biochar with longer burn times.

HISTORY OF BIOCHAR

The name 'biochar' was coined in 2005, although 'charcoal' has been used for agricultural purposes as a soil amendment for thousands of years. Improved nutrient retention and nutrient availability for plants in Amazonian soils have been linked to the presence of char in the soil profile (Glaser et al., 2002).

These soils, within the Amazon River basin, are carbon rich and it is believed that a form of biochar or charcoal had been used in agriculture. These dark soils, known as '*terra preta*', were likely to originate from charred organic materials like manure, crop residue and bones that were set alight and covered with dirt to eliminate oxygen but hold in the heat from the fire which, in turn, baked the organic matter.

Carbon-containing materials such as plants are made from carbon that is sourced from the air. While alive and whole, that organic matter holds the carbon in place, but when it burns or decomposes the carbon is released back into the atmosphere in gases like carbon dioxide. These gases then contribute to climate change and global warming.

If the organic material undergoes pyrolysis, then the carbon largely remains locked as a solid in the biochar and can remain stable. Because the half-life of the carbon found in biochar is greater than 1000 years, biochar, when applied to soil, is a carbon sequestration tool (Laird, 2008). Other by-products of the pyrolysis process include oils and methane that can be captured and used as an environmentally friendly fuel.

BIOCHAR RESEARCH REVIEW

Although there has been little research done on the long-term effects of biochar used to amend sports turf profiles, there has been extensive research into biochar's benefits on agricultural soils around the world. The following is a review of some of the research that has been conducted.

SOIL REMEDIATION

According to Beesley (2011), who reviewed the role of biochar in the remediation of contaminated soils in a number of studies, it was concluded that biochar can effectively remove persistent organic pollutants and some pesticides from soils.

Hilber et al. (2017) reviewed many papers on biochar and its effects on persistent organic pollutants (POPs) and heavy metals in soils and concluded that due to their high surface area, high CEC and persistence in the soil, heavy metals precipitate as phosphates on the surface of biochar due to the rise in pH (e.g. copper, zinc, lead and cadmium). However, metalloids and non-cationic negatively charged metals such as arsenic aren't precipitated out on biochar surfaces.